RAG Learning

Topic 2: Landscape Systems, Processes & Change — Coastal Landscapes

2.1 The Coast as a System

Littoral zone: The area between the highest and lowest points affected by wave action, including the coast and nearshore zone. Coasts operate as *open systems* with inputs (energy, sediment), outputs (sediment removal, energy dissipation), stores (beaches, dunes, mudflats), and transfers (longshore drift, erosion, deposition).

Systems Approach

Component	Examples
Inputs	Wave energy (determined by wind speed, duration, fetch), tidal energy, wind energy, sediment (from rivers, cliff erosion, offshore sources, biological material), solar energy, precipitation, geological structure
Outputs	Sediment lost offshore or alongshore beyond the cell boundary, dissolved minerals, energy dissipation through wave breaking, evaporation
Stores	Beaches (sand, shingle), sand dunes, spits, bars, tombolos, mudflats, salt marshes, cliff faces, offshore bars
Transfers/Flows	Longshore drift, onshore/offshore sediment transport, erosion processes, weathering, mass movement, deposition, aeolian (wind) transport

Sediment cell (littoral cell): A self-contained stretch of coastline within which sediment movement is largely contained in a cycle of sources, transfers, and sinks. England and Wales have 11 major

sediment cells. Sediment rarely crosses cell boundaries (marked by headlands or estuary mouths). They form the basis for Shoreline Management Plans (SMPs). Understanding the sediment budget within each cell is crucial for effective coastal management.

Dynamic equilibrium: A balanced state where the coast maintains a balance between inputs and outputs of energy and sediment over time. The coast fluctuates around this balance naturally (seasonal changes in wave energy, storms) but returns to equilibrium. Human intervention (e.g. building groynes, constructing sea walls, damming rivers) disrupts the equilibrium, causing **feedback effects** elsewhere in the system — often accelerating erosion downdrift.

Feedback Loops in Coastal Systems

- **Negative feedback (self-regulating):** Cliff collapse adds sediment to the beach → wider beach absorbs more wave energy → reduces wave impact on cliff → slows further erosion. The system tends to return towards equilibrium.
- **Positive feedback (self-reinforcing):** Sea-level rise increases wave energy reaching the cliff base → accelerated erosion → narrower beach (sediment removed) → less wave energy absorption → even more erosion. The system moves further from equilibrium. Also triggered by human actions: groynes trap sediment updrift → downdrift beach starved → increased erosion → narrower beach → more erosion (terminal groyne syndrome).

2.2 Waves

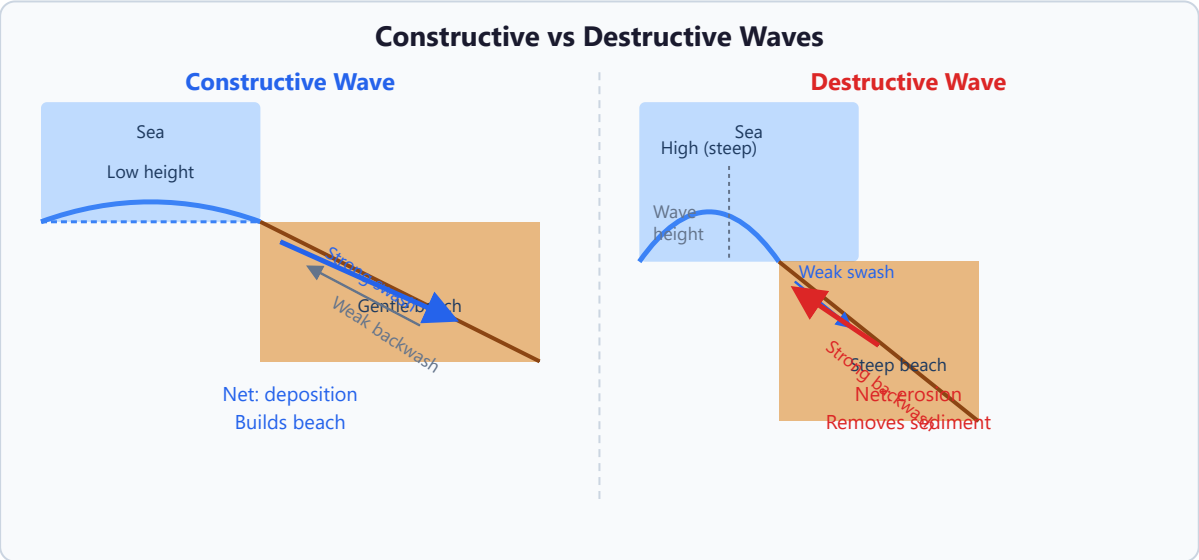
Wave: An oscillation of water caused by wind blowing over the sea surface. Energy is transferred through the water but the water itself does not move forward (particles follow circular orbits in deep water). Wave energy is determined by wind speed, wind duration, and fetch. Wave energy is proportional to the square of wave height — doubling height quadruples energy.

Constructive vs Destructive Waves

Feature	Constructive	Destructive
Frequency	6–8 per minute (low)	10–14 per minute (high)

Wave height	Low (<1 m)	High (>1 m, often 3–5 m in storms)
Wavelength	Long	Short
Swash vs backwash	Strong swash, weak backwash (net onshore movement)	Weak swash, strong backwash (net offshore movement)
Effect	Deposits sediment, builds beaches, creates berms	Erodes beaches, removes sediment, creates offshore bars
Wave steepness	Low (spilling breakers)	High (plunging breakers)
Origin	Distant storms — swell waves travel thousands of km	Local storms — wind-generated waves with short fetch

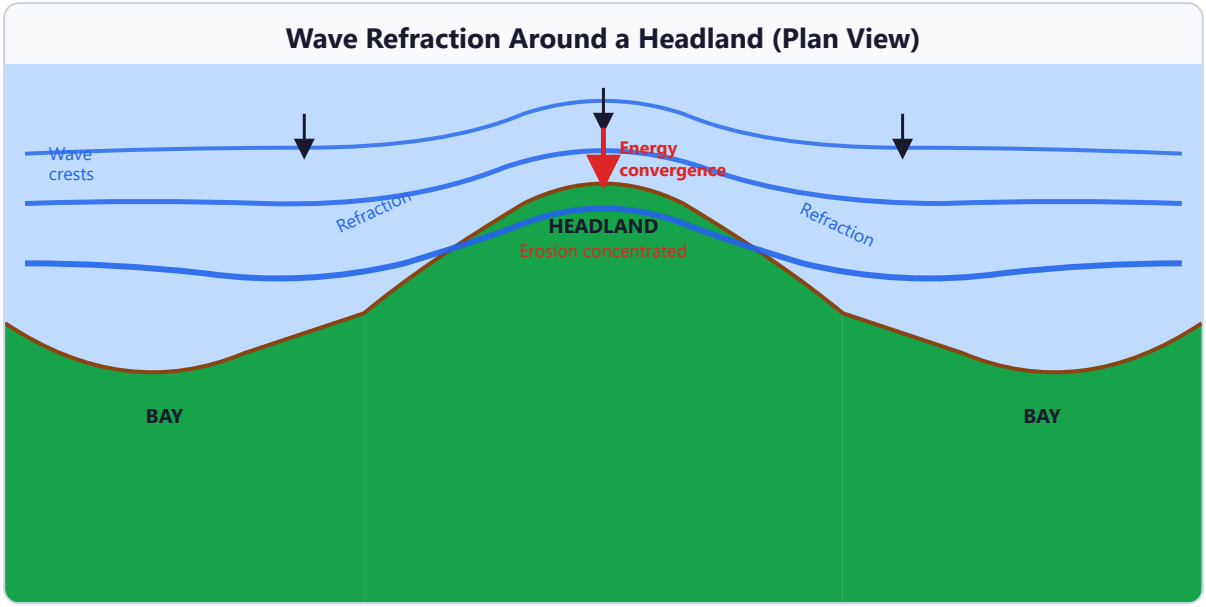
Seasonal beach profiles: In summer, calmer conditions produce constructive waves that build wide, gently-sloping beaches with prominent berms. In winter, storm waves (destructive) strip material from the upper beach and deposit it offshore as breakpoint bars. The beach steepens and narrows. This seasonal cycle demonstrates dynamic equilibrium.



Wave Refraction

When waves approach an irregular coastline, they slow in shallow water near headlands but continue at speed in deeper water in bays. This causes wave fronts to curve (refract), concentrating energy on headlands (causing erosion) and dispersing energy in bays (encouraging deposition). Over geological time,

wave refraction tends to straighten coastlines — but never completely, because the geology that creates differential resistance persists.



2.3 Coastal Processes

Marine Erosion

- **Hydraulic action** — the force of water and trapped air compressed into cracks and joints. Pressures can reach 30 tonnes/m^2 during storms. Air is compressed, then expands explosively as the wave retreats, widening fractures. The dominant erosion process on cliffs with well-developed joint systems.
- **Abrasion (corrasion)** — rock fragments carried by waves scrape and wear away the cliff face, acting like sandpaper. Most effective during storms when large pebbles and cobbles are hurled at cliffs. Creates smooth, polished rock surfaces and enlarges wave-cut notches.
- **Attrition** — rock fragments carried by the sea collide with each other, becoming progressively smaller, smoother, and more rounded. Explains the reduction in sediment size along a beach in the direction of longshore drift.
- **Corrosion (solution)** — weak carbonic acid in seawater dissolves calcium carbonate in limestone and chalk. Enhanced in warmer water and where biological activity (organisms) increases acidity. Creates distinctive features on limestone coasts: blowholes, narrow geos, and sculpted surfaces.

Sub-aerial Processes

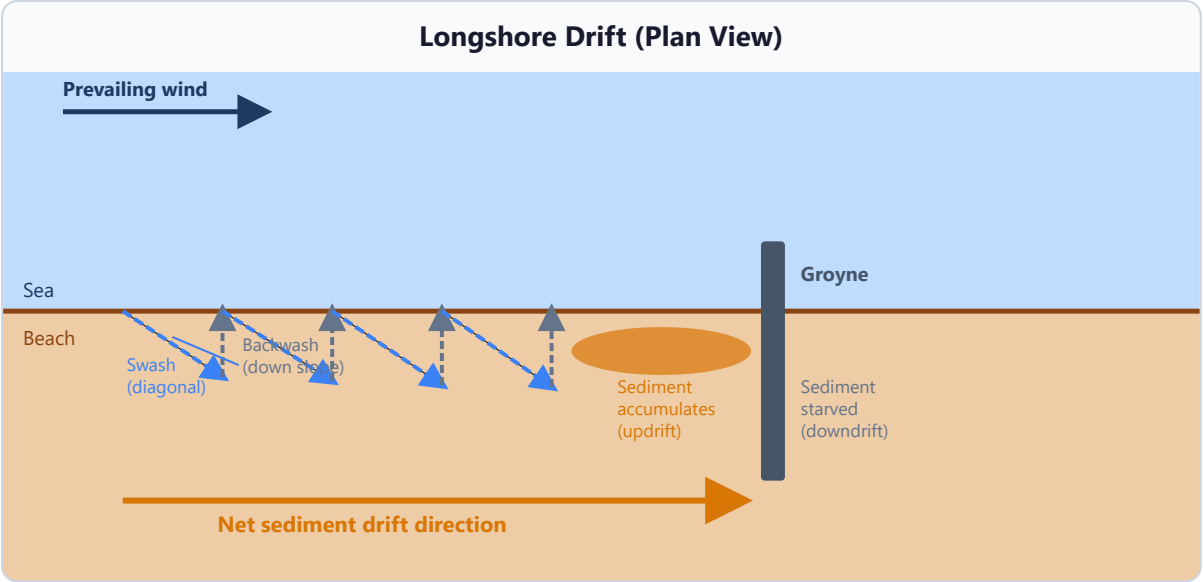
- **Weathering** — the in-situ breakdown of rock at or near the surface:

- *Mechanical:* freeze-thaw (water in cracks expands by 9% on freezing, exerting pressures up to 2,100 kg/cm^2); salt crystallisation (salt spray dries in cracks and pores, crystals grow and exert pressure, particularly effective on sandstone); wetting and drying (clay minerals swell when wet and contract when dry, causing crumbling)
- *Chemical:* carbonation ($CO_2 + H_2O \rightarrow H_2CO_3$ dissolves $CaCO_3$ in limestone/chalk); oxidation (iron minerals react with oxygen, weakening rock); hydrolysis (water reacts with feldspar minerals to form clay)
- *Biological:* plant roots widen cracks (tree roots can exert pressures of 1–15 kg/cm^2); burrowing organisms (piddock molluscs bore into soft rock); organic acids from lichens dissolve rock surfaces

- **Mass movement** — downslope movement under gravity, triggered by wave undercutting, rainfall saturation, or loading:
  - *Rockfall:* sudden collapse of individual rocks or blocks from steep/vertical cliff faces. Common on well-jointed hard rock (chalk, limestone). Very fast. Dangerous to walkers below.
  - *Landslide:* coherent block of material slides along a planar failure surface. Occurs where permeable rock (e.g. chalk, sandstone) overlies impermeable rock (e.g. clay), creating a slip plane when the contact layer becomes saturated.
  - *Rotational slump:* material slides along a curved, concave failure surface, rotating backwards. Characteristic of clay cliffs (e.g. London Clay at Isle of Sheppey, Barton-on-Sea). Often leaves a stepped profile with "slump scars".
  - *Mudflow:* saturated fine-grained material flows downslope as a viscous mass. Very rapid. Triggered by prolonged heavy rainfall on clay or silt slopes.
  - *Soil creep:* imperceptibly slow movement of soil particles downslope. Causes tilted fences, leaning walls, and curved tree trunks. Continuous process measured in mm/yr.

Sediment Transport: Longshore Drift

Longshore drift is the dominant process moving sediment along the coast. Waves approach at an angle (determined by the prevailing wind direction), carrying sediment diagonally up the beach with the swash. Backwash pulls sediment straight back down under gravity. The net result is a zig-zag movement along the shore. In southern England, the prevailing SW winds drive drift west to east along the south coast and south to north up the east coast. Evidence of longshore drift: sediment accumulation on the updrift side of groynes; formation of spits at changes in coastline direction; progressive decrease in sediment size in the drift direction (attrition).



Other transport mechanisms: traction (large particles rolled along seabed), saltation (pebbles bounced), suspension (fine silt/clay held in water column), solution (dissolved minerals).

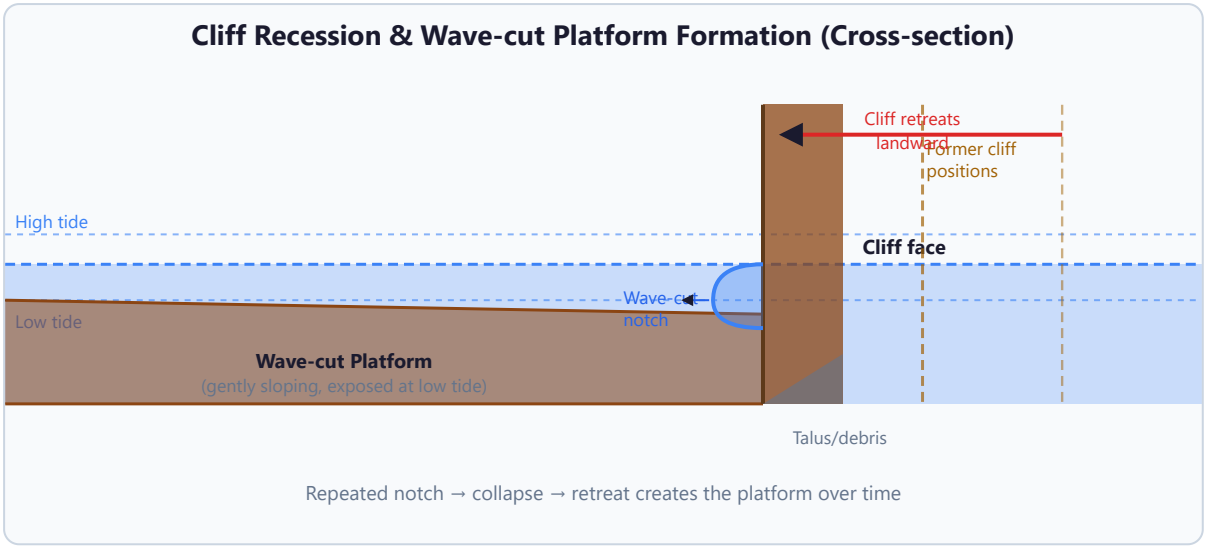
2.4 Erosional Landforms

Headlands and Bays

Form on **discordant coastlines** where alternating bands of resistant and less resistant rock are perpendicular to the shore. Differential erosion creates bays in soft rock (clay, sands) and headlands in hard rock (granite, limestone, chalk). On **concordant coastlines** (rock bands parallel to coast), initial breaching of the resistant outer band allows rapid erosion of softer rock behind, potentially forming coves (e.g. Lulworth Cove, Dorset).

Cliff Recession and Wave-cut Platforms

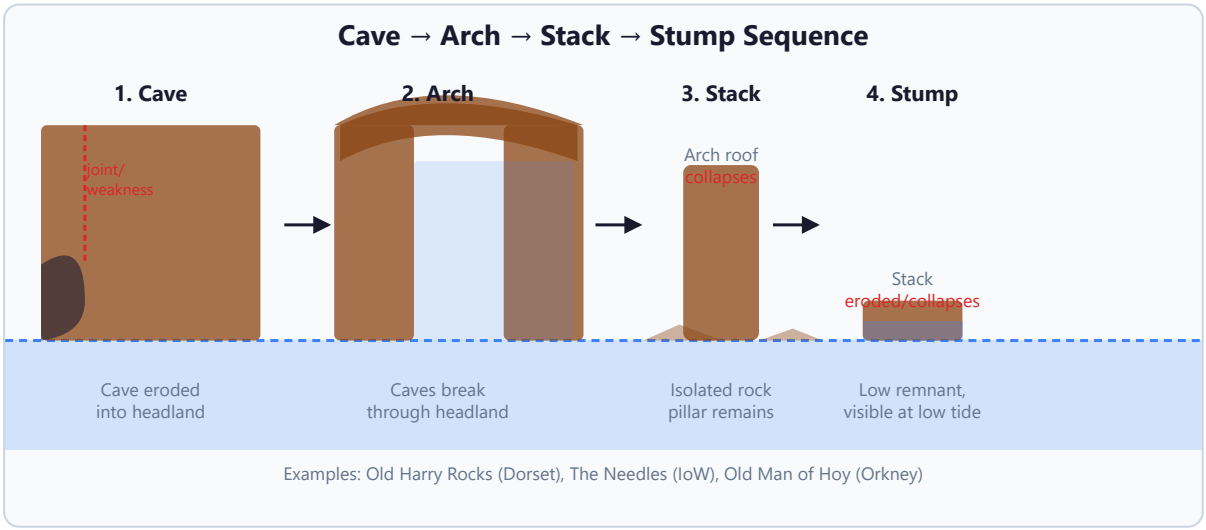
Waves erode a **wave-cut notch** at the cliff base between high and low tide marks. As the notch deepens, the unsupported cliff above collapses. Repeated cycles cause the cliff to **retreat landward**, leaving a gently sloping (1–4°) **wave-cut platform** exposed at low tide. Platform width eventually limits further erosion by dissipating wave energy before it reaches the cliff. UK chalk cliff recession rates: Beachy Head ~0.5 m/yr; Holderness boulder clay cliffs: ~1.8 m/yr (up to 10 m in single storm events).



Caves, Arches, Stacks, and Stumps

A sequence of erosional features developing on headlands:

1. **Cave** — waves exploit a joint, fault, or weakness through hydraulic action and abrasion, eroding a hollow into the cliff face.
2. **Arch** — two caves on opposite sides of a narrow headland break through to form a natural arch. The arch roof is attacked from above by weathering and below by wave erosion.
3. **Stack** — the arch roof collapses, leaving an isolated pillar of rock. Examples: Old Harry Rocks (Dorset chalk), The Needles (Isle of Wight chalk), Orkney sea stacks (Old Man of Hoy — sandstone, 137 m tall).
4. **Stump** — the stack is undercut and collapsed, leaving a low remnant often only visible at low tide.



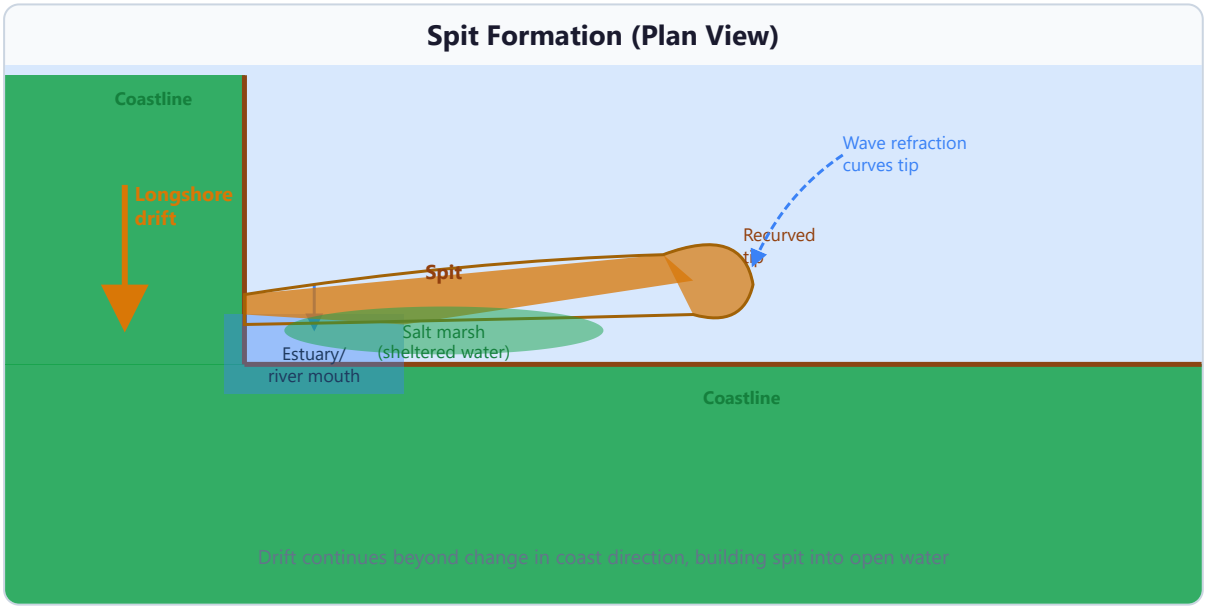
2.5 Depositional Landforms

Beaches

Beaches form wherever sediment supply exceeds removal. Sandy beaches: wide, gently sloping (1–5°), fine material packs together. Shingle beaches: steeper (10–20°), more permeable (backwash infiltrates, weakening return flow). Beach features: **berms** (ridges marking high tide deposits), **cusps** (crescent-shaped hollows), **runnels** (shallow drainage channels), **ripples** (small undulations from wave action).

Spits, Bars, and Tombolos

- **Spit:** elongated finger of sand/shingle extending from the coast where the shoreline changes direction or at a river mouth. Formed by longshore drift continuing beyond the change in coastline. Recurved tip from wave refraction. Salt marsh often develops in sheltered water behind the spit. E.g. Spurn Head (5.5 km, Humber estuary), Hurst Castle Spit (2 km, Hampshire).
- **Bar:** a spit that grows completely across a bay, enclosing a lagoon. E.g. Slapton Ley (Devon) — freshwater lagoon behind shingle bar.
- **Tombolo:** a bar connecting an island to the mainland. E.g. Chesil Beach (28 km, connecting Isle of Portland to Dorset), St Ninian's Tombolo (Shetland).

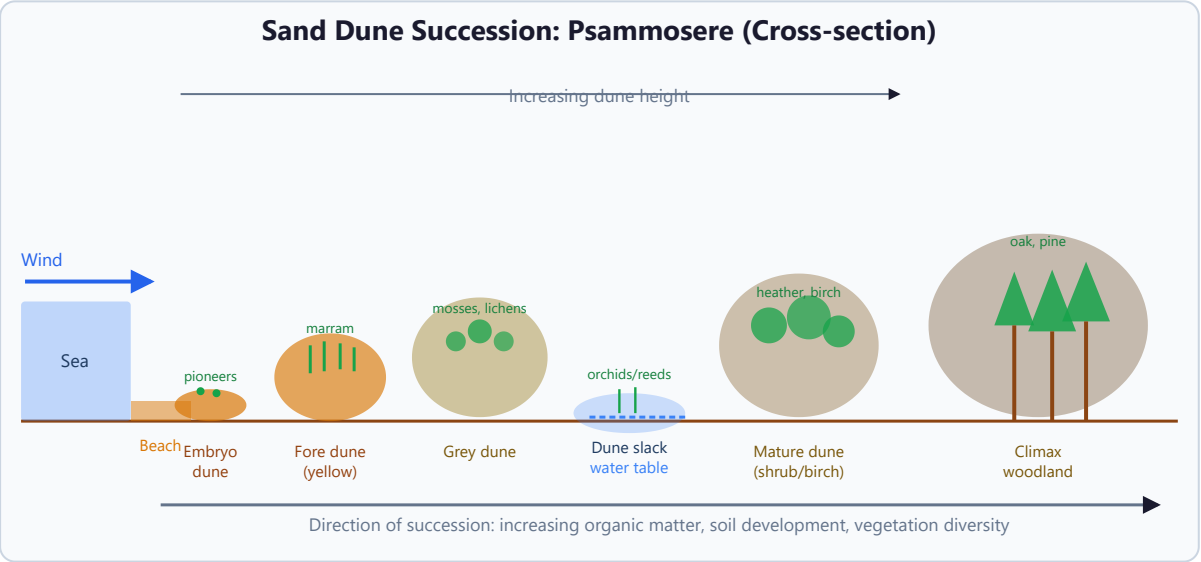


Sand Dunes

Form where onshore winds blow dry sand inland from a wide beach. A succession (**psammosere**) develops from embryo dunes to climax woodland:

Zone	Height	Vegetation	Soil/Environment

Embryo dunes	0.5–2 m	Pioneer species: sea rocket, sand couch grass	~0% organic matter, alkaline (pH 8+, shell fragments), salty, dry, mobile
Fore dunes (yellow)	5–10 m	Marram grass (deep roots bind sand, tolerates burial)	1–2% organic, pH 7–8, still mobile, some humus developing
Grey dunes	8–12 m	Greater diversity: mosses, lichens, wildflowers, heather	5–10% organic, pH 5–7, more stable, visible humus layer
Dune slacks	Low (between ridges)	Wetland species: creeping willow, orchids, reeds	Water table at or near surface, unique habitat
Mature/climax	Stable	Shrubs, birch, pine, oak woodland	Well-developed soil, pH 4–5 (acidic), thick humus



Salt Marshes

Develop in sheltered, low-energy environments (behind spits, in estuaries). Fine sediment (silt, mud) deposited by tides. Pioneer plants (glasswort/Salicornia, cord grass/Spartina) colonise, trapping more sediment and raising the surface. A **halosere** succession develops: pioneer zone (daily inundation) → lower marsh → upper marsh → transitional. Salt marshes are ecologically vital: nursery habitats for fish and invertebrates, feeding grounds for migratory birds, carbon sinks (sequestering carbon at 2–4x the rate of tropical forests per hectare), and natural flood defences (absorbing wave energy and tidal surge).

2.6 Sea Level Change and Coastal Landscapes

Eustatic change: Global change in sea level caused by changes in ocean water volume (glacial melting/formation) or ocean basin shape. During the Last Glacial Maximum (~20,000 years ago), sea level was ~120 m lower than today. Since 1900, global mean sea level has risen ~20 cm; currently rising at ~3.6 mm/yr (accelerating). Causes: thermal expansion (~40%), glacier/ice cap melt (~30%), Greenland/Antarctic ice sheet melt (~30%).

Isostatic change: Local/regional change in the relative level of land and sea. During glaciation, the weight of ice depresses the land; after deglaciation, the land slowly rebounds (post-glacial isostatic rebound). In the UK: Scotland rising ~1–3 mm/yr; SE England sinking ~1–2 mm/yr (relative sea-level rise is therefore faster in SE England).

Coastal Landforms of Sea Level Change

- **Emergent features (land rising / sea falling):** raised beaches (former beaches now above current sea level), relict cliffs (vegetated, no longer eroded), elevated wave-cut platforms. E.g. raised beaches at 8–10 m in western Scotland.
- **Submergent features (sea rising / land sinking):** rias (drowned river valleys — e.g. Kingsbridge Estuary, Devon), fjords (drowned glacial valleys with steep sides and shallow threshold — e.g. Sognefjorden, Norway: 204 km long, 1,308 m deep), Dalmatian coastlines (drowned valleys parallel to coast creating elongated islands — Croatian coast).

2.7 Coastal Management

Shoreline Management Plans (SMPs)

Strategic plans covering entire sediment cells, setting management policy for each stretch of coast over 20, 50, and 100-year timescales. Four policy options:

- **Hold the Line** — maintain existing defences, building new ones where necessary. Appropriate where high-value assets (towns, infrastructure) are at risk.
- **Advance the Line** — build new defences seaward of existing ones, reclaiming land from the sea. Rare and expensive.

- **Managed Realignment (Managed Retreat)** — allow controlled flooding of low-value land to create new intertidal habitats, moving defences inland. Cost-effective and environmentally beneficial.
- **No Active Intervention** — allow natural processes to operate without human interference. Applied where the coast has low economic/social value or where intervention is not cost-effective.

Hard vs Soft Engineering

Hard Engineering	Cost	Advantages	Disadvantages
Sea wall	£5,000–10,000/m	Effective barrier, long-lasting (50+ yrs), public reassurance	Very expensive, reflects wave energy (scouring beach at base), visually intrusive, may need constant repair
Groynes	£5,000–10,000 each	Traps sediment, builds beach (natural defence), relatively cheap	Starves downdrift areas of sediment (terminal groyne syndrome), needs replacement (wood: 20 yrs, rock: 50+ yrs)
Rock armour (rip-rap)	£1,000–3,000/m	Absorbs wave energy, relatively cheap, quick to install	Unattractive, can shift in storms, does not address cause of erosion
Offshore breakwaters	£5,000+/m	Reduces wave energy before reaching shore, creates calmer water for deposition	Expensive, interferes with navigation, can create unexpected sediment patterns, visual impact
Revetments	£2,000–5,000/m	Absorbs/dissipates wave energy, protects cliff base	Can be undermined, limited lifespan, needs maintenance

Soft Engineering	Cost	Advantages	Disadvantages
Beach nourishment	£3,000–5,000/m	Natural appearance, absorbs wave energy, recreational value, maintains sediment budget	Temporary (replenishment every 5–10 years), ongoing cost, dredged material may differ from natural sediment

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Managed realignment	Variable (often cheaper long-term)	Creates habitats (salt marsh, mudflats), natural flood defence, sustainable, biodiversity gains, carbon sequestration	Loss of farmland, compensation to landowners, local opposition, slow to establish, requires long-term commitment
Dune stabilisation	Low	Natural defence, biodiversity, recreational value, cheap	Vulnerable to storms, requires visitor management, slow to establish
Cliff drainage	Moderate	Reduces mass movement risk by lowering water table in cliff	Does not prevent marine erosion at base, requires maintenance

Case Study: Holderness Coast, East Yorkshire — Fastest Eroding Coast in Europe

The Holderness coast stretches 61 km from Flamborough Head (resistant chalk headland) to Spurn Head (sand/shingle spit). The cliffs are composed of glacial till (boulder clay) deposited during the last ice age — soft, unconsolidated, and easily eroded. Average retreat: 1.8 m/yr (some sections up to 10 m in single storms). Since Roman times, ~4 km of land has been lost, and 30+ villages have disappeared (e.g. Ravenser Odd, a medieval port that once rivalled Hull). **Why so fast?** Weak geology (boulder clay crumbles when saturated), powerful North Sea waves (long fetch from NNE, storm surges), narrow beaches (clay produces little sand/gravel, so natural beach protection is minimal), rising sea levels. **Management conflicts:** Mappleton received £2 million rock armour and groynes (1991), protecting 50 properties — but starved beaches to the south, accelerating erosion at Cowden and Aldbrough. Bridlington and Withernsea have sea walls and groynes. Spurn Head: No Active Intervention (left to natural processes — breached in 2013 tidal surge, road severed). The Holderness coast exemplifies the coastal management dilemma: protecting one area disrupts the sediment budget and accelerates erosion elsewhere. The SMP designates most of the coast as "No Active Intervention" except at settlements, causing resentment among farming communities losing land.

Case Study: Medmerry, West Sussex — Managed Realignment

The UK's largest open-coast managed realignment scheme, completed by the Environment Agency in 2013. A new 7 km earth embankment was built 500 m inland, and the old shingle bank was deliberately breached, allowing the sea to flood 183 hectares of low-lying farmland. **Cost:** £28 million (compared to an estimated £300+ million to maintain traditional hard defences over 100 years — a 10:1 cost saving). **Benefits:** 348 properties now better protected from coastal flooding, 183 ha of new

intertidal habitat created (salt marsh, saline lagoons), supporting avocets, little terns, water voles, and rare aquatic invertebrates; natural flood defence (salt marsh absorbs wave energy); carbon sequestration; RSPB manages as a nature reserve attracting 50,000+ visitors/yr. **Challenges:** loss of productive farmland, disruption during construction, some local opposition, long-term maintenance of the new embankment. Medmerry demonstrates that managed realignment can deliver flood protection, biodiversity gains, carbon storage, and cost savings simultaneously — a "win-win-win" approach. It is increasingly seen as a model for future coastal management as sea levels rise.

Case Study: The Dorset Coast (Jurassic Coast) — Geological Diversity and Landforms

The Dorset coast (part of the 155 km Jurassic Coast World Heritage Site, designated 2001) displays an exceptional range of landforms due to its diverse geology spanning 185 million years. **Lulworth Cove:** classic concordant coastline feature. Waves breached the resistant Portland Limestone outer band and rapidly eroded the softer Wealden Clay behind, creating a near-circular cove 450 m wide. **Durdle Door:** natural limestone arch formed by marine erosion exploiting a weakness in the Portland Limestone headland. **Old Harry Rocks:** chalk stacks and stumps at Handfast Point, demonstrating the cave-arch-stack-stump erosional sequence. **Chesil Beach:** 28 km shingle tombolo connecting the Isle of Portland to the mainland, with sediment graded from pea-sized pebbles at the west end to cobble-sized at the east end (due to longshore drift sorting and wave energy differences). The Fleet Lagoon behind Chesil Beach is the largest tidal lagoon in the UK. The Dorset coast illustrates how geology controls coastline shape: concordant and discordant sections produce contrasting landforms, and lithology (rock type) determines erosion rates and cliff profiles.

Exam Tip: Edexcel frequently asks you to evaluate coastal management strategies using concepts of sustainability. The most sustainable approaches work WITH natural processes rather than against them. Managed realignment and beach nourishment maintain sediment budgets; hard engineering disrupts them. Frame your evaluation around: economic sustainability (cost-effectiveness over time), environmental sustainability (habitat creation vs destruction), and social sustainability (who benefits, who loses).

Exam Tip: For Edexcel, always think in terms of SYSTEMS. A groyne does not just trap sediment — it disrupts the sediment budget of the entire littoral cell, creating positive feedback loops downdrift. A managed realignment does not just flood farmland — it restores natural equilibrium by allowing the

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coast to adjust to rising sea levels. Systems thinking elevates your answers from descriptive to analytical.

Exam Tip: When discussing landforms, always link PROCESS to FORM. Do not just describe what a spit looks like — explain HOW longshore drift creates it, WHY it curves at the tip, and HOW salt marshes form in the sheltered water behind it. Every landform is the product of specific processes operating in specific conditions. This process-form linkage is the core of physical geography.

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